# Ground Track of GPS Satellite PRN-18 Computed from YUMA Almanac Keplerian Elements

Oscar Gonzalo Aran and Adam Ruiz Herrero
*Navegació Aèria, Cartografia i Cosmografia (NACC) — Lab. Session 1, Group 18*
*Escola d'Enginyeria de Telecomunicació i Aeroespacial de Castelldefels (EETAC), Universitat Politècnica de Catalunya (UPC)*
*October 2026*

***Abstract*—This report presents the MATLAB implementation of a complete procedure to compute the sub-satellite point and the ground track of a GPS satellite from the Keplerian elements broadcast in the YUMA almanac. The target UTC epoch is converted into GPS time, the mean anomaly is propagated from the almanac Time of Applicability, Kepler's equation is solved with a Newton–Raphson iteration, and the resulting in-plane position is rotated into the Earth-Centred Earth-Fixed frame, accounting for Earth rotation and for the drift of the ascending node. The position is finally converted into WGS84 geodetic coordinates. The method is applied to PRN-18 with the week-390 almanac (ToA = 405 504 s) for the epoch 1 October 2026, 10:21:07 UTC. The satellite is found at 150.65° W, 47.30° N, at an altitude of 20 317 km. Propagating the orbit in 288 steps of 5 min yields the characteristic two-lobe GPS ground track, which reaches ±55.6° of latitude and repeats after two orbital periods (≈ 86 156 s). The result is compared with the FAA NSTB real-time satellite status tool and cross-checked against an earlier almanac. The cross-check shows that the gravitational parameter adopted in the code ($\mu = G \cdot M_e$) introduces an along-track error that grows linearly with the propagation interval, which makes it advisable to use the most recent almanac.**

***Index Terms*—GPS, YUMA almanac, Keplerian elements, Kepler's equation, ECEF, geodetic coordinates, sub-satellite point, ground track.**

## I. Introduction

The *ground track* of a satellite is the locus of its *sub-satellite point*, i.e., the intersection of the Earth's surface with the line joining the satellite and the Earth's centre (or, in geodetic terms, the point of the ellipsoid below the satellite with zero altitude). Ground tracks are the basic tool to visualise the coverage and visibility of a satellite and are routinely used in mission and navigation planning.

The GPS space segment consists of satellites in medium Earth orbit, at an altitude of about 20 200 km, distributed in six orbital planes inclined approximately 55° with respect to the equator. Their orbital period is half a sidereal day, so each satellite completes two revolutions while the Earth rotates once with respect to the orbital plane, and its ground track repeats every day [10]. Every GPS satellite broadcasts an *almanac*: a reduced set of Keplerian elements of the whole constellation, valid for several days, which receivers use to predict which satellites are in view. The U.S. Coast Guard Navigation Center and CelesTrak publish these almanacs in the plain-text YUMA format [3]–[5].

The objective of this lab. session [1] is to develop, following a prescribed code structure, a set of MATLAB functions that compute the sub-satellite point of a GPS satellite at any epoch from its YUMA almanac, and to use them to plot its complete ground track. The satellite assigned to group 18 is PRN-18. Section II describes the input data, Section III the method and formulas used in each of the nine steps of the statement, Section IV the results and their validation, and Sections V and VI the discussion and conclusions.

## II. Input Data

### A. YUMA Almanac of PRN-18

Calling *gpstime()* for the target epoch downloads from CelesTrak the almanac whose ToA is the closest valid one, in this case *almanac_week_0390.405504.txt* (GPS week 390, ToA = 405 504 s, i.e. Thursday 1 October 2026, 16:38:24 GPS time). The record of PRN-18 was copied from this file into *main1.m* (Table I). The almanac uses the semi-major axis through its square root, so $a = (\sqrt{A})^2 = 26\,559\,161$ m. The clock parameters $a_{f0}$ and $a_{f1}$ are not needed to compute the position.

TABLE I
WEEK-390 ALMANAC OF PRN-18 (TOA = 405 504 S)

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| ID | PRN number | 18 | — |
| Health | SV status (000 = usable) | 000 | — |
| $e$ | Eccentricity | $6.573200226 \times 10^{-3}$ | — |
| $t_{oa}$ | Time of applicability | 405 504 | s |
| $i_0$ | Orbital inclination | 0.9699576238 | rad |
| $\dot{\Omega}$ | Rate of right ascension | $-8.068907531 \times 10^{-9}$ | rad/s |
| $\sqrt{A}$ | Sqrt. of semi-major axis | 5153.558105 | $m^{1/2}$ |
| $L_o$ | AN long. at week epoch | −0.6772630040 | rad |
| $\omega$ | Argument of perigee | −2.812811435 | rad |
| $M_0$ | Mean anomaly at ToA | 0.9359263948 | rad |
| $a_{f0}$ | SV clock bias (not used) | $-2.431869507 \times 10^{-4}$ | s |
| $a_{f1}$ | SV clock drift (not used) | $7.275957614 \times 10^{-12}$ | s/s |
| WN | GPS week (modulo 1024) | 390 | — |

### B. Constants and Target Epoch

The physical constants imposed by the statement are listed in Table II, together with the WGS84 ellipsoid parameters used in the geodetic conversion. The target epoch is **1 October 2026, 10:21:07 UTC** (12:21:07 local summertime in Spain, UTC+2), the instant at which the FAA reference screen capture was taken.

TABLE II
CONSTANTS AND PARAMETERS

| Symbol | Parameter | Value |
|---|---|---|
| $G$ | Gravitational constant | $6.67384 \times 10^{-11}$ m³ kg⁻¹ s⁻² |
| $M_e$ | Earth mass | $5.972 \times 10^{24}$ kg |
| $\mu$ | Grav. parameter, $G \cdot M_e$ | $3.985617 \times 10^{14}$ m³/s² |
| $\omega_e$ | Earth rotation rate | $7.2921151467 \times 10^{-5}$ rad/s |
| tol | Kepler tolerance | $10^{-8}$ rad |
| $\Delta t_{LS}$ | Leap seconds GPS − UTC | 18 s |
| $a_E$ | WGS84 semi-major axis | 6 378 137 m |
| $f$ | WGS84 flattening | 1/298.257223563 |
| $\Delta t$ | Ground-track time step | 300 s |

## III. METHODOLOGY

The computation is split into the functions prescribed in [1]. Every function writes its intermediate result to the debugging file *main1.dat* when the global file descriptor *fd* exists. The formulas follow the course slides [2] and are consistent with the GPS interface specification [6].

### *A. Step 1 – GPS Time, Elapsed Time and Node Longitude at ToA*

The provided function *gpstime()* converts the UTC date into a Julian Date (JD) with the algorithm of [9] (with $y$ = YYYY, $m$ = MM if MM > 2, and $y$ = YYYY − 1, $m$ = MM + 12 otherwise):

$$JD = \lfloor 365.25\, y \rfloor + \lfloor 30.6001(m+1) \rfloor + D + \frac{UT}{24} + 1\,720\,981.5 \quad (1)$$

where $\lfloor \cdot \rfloor$ denotes the integer part and UT is the time of day in hours. The GPS time in days since the GPS epoch (6 January 1980, JD 2 444 244.5), the GPS week number WN and the seconds elapsed since the beginning of the week (*esec*) are

$$t_{GPS} = JD - 2\,444\,244.5 + \frac{\Delta t_{LS}}{86400} \quad (2)$$

$$WN = \lfloor \frac{t_{GPS}}{7} \rfloor,\ esec = 86400(t_{GPS} - 7\,WN) \quad (3)$$

For the target epoch, JD = 2 461 314.931331, WN = 2438 (390 + 2 roll-overs of 1024 weeks) and *esec* = 382 885 s (Thursday 10:21:25 GPS time, 18 s ahead of UTC). The time elapsed from the ToA to the target epoch is

$$dt = esec - t_{oa} \quad (4)$$

which gives $dt$ = −22 619 s: the target epoch is 6 h 17 min *before* the ToA, so the orbit is propagated backwards. The almanac parameter $L_o$ is the longitude of the ascending node (AN) at the week epoch. Since the AN is fixed in inertial space while the Earth rotates eastwards, its longitude decreases at the rate $\omega_e$, so the longitude of the AN at the ToA is

$$\Omega_0 = L_o - \omega_e t_{oa} \quad (5)$$

which yields $\Omega_0$ = 1.168845 rad (modulo 2π).

### *B. Step 2 – Mean Anomaly*

The mean anomaly grows linearly with time at the mean motion $n$, obtained from Kepler's third law:

$$n = \sqrt{\frac{\mu}{a^3}},\ M = M_0 + n\,dt \quad (6)$$

With $\mu = G\,M_e$, $n$ = 1.458567 × 10⁻⁴ rad/s and $M$ = 3.919980 rad (modulo 2π) at the target epoch.

### *C. Step 3 – Eccentric Anomaly (Kepler's Equation)*

The eccentric anomaly $E$ is related to $M$ by Kepler's equation,

$$M = E - e \sin E \quad (7)$$

which is transcendental and is solved iteratively with the Newton–Raphson method, starting from $E_0 = M$:

$$E_{k+1} = E_k - \frac{E_k - e \sin E_k - M}{1 - e \cos E_k} \quad (8)$$

The loop stops when the correction $|E_{k+1} - E_k|$ falls below tol = $10^{-8}$ rad. Because GPS orbits are almost circular ($e \approx 0.0066$), convergence is reached in two or three iterations: at the target epoch the corrections are $4.59 \times 10^{-3}$, $4.84 \times 10^{-8}$ and $< 10^{-15}$ rad, giving $E$ = 3.915386 rad. Substituting $E$ back into (8) reproduces $M$ with a residual below $2 \times 10^{-15}$ rad for all 289 epochs.

### *D. Step 4 – True Anomaly*

The true anomaly ν follows from $\tan(\nu/2) = \sqrt{((1+e)/(1-e))}\,\tan(E/2)$. To obtain the correct quadrant the two-argument arctangent is used:

$$\nu = 2\ atan2\left(\sqrt{1+e}\,\sin\frac{E}{2}, \sqrt{1-e}\,\cos\frac{E}{2}\right) \quad (9)$$

giving ν = 3.910803 rad.

### *E. Step 5 – Coordinates in the Orbital Plane*

The orbital radius and the argument of latitude $u$ (angle from the AN to the satellite) are

$$r = a(1 - e\cos E),\ u = \omega + \nu \quad (10)$$

and, in a frame lying on the orbital plane with its $x$ axis pointing to the AN,

$$x_p = r\cos u,\ y_p = r \sin u \quad (11)$$

At the target epoch, $r$ = 26 684 031 m, $u$ = 62.91°, $x_p$ = 12 151 505.6 m and $y_p$ = 23 756 650.2 m.

### *F. Step 6 – ECEF Coordinates*

The longitude of the AN at the ToA must be corrected for the slow drift of the node ($\dot{\Omega} < 0$, a westward regression mainly caused by the Earth's oblateness) and for the Earth rotation, both proportional to $dt$. The current longitude of the AN is therefore

$$\Omega = \Omega_0 + (\dot{\Omega} - \omega_e)\,dt \quad (12)$$

The ECEF coordinates are obtained by rotating the orbital-plane vector by the inclination $i_0$ around the AN line and by Ω around the polar axis, $[x\ y\ z]^T = R_3(-\Omega)\,R_1(-i_0)\,[x_p\ y_p\ 0]^T$:

$$x = x_p\cos\Omega - y_p\cos i_0 \sin\Omega \quad (13)$$

$$y = x_p\sin\Omega + y_p\cos i_0\cos\Omega \quad (14)$$

$$z = y_p \sin i_0 \quad (15)$$

At the target epoch Ω = 2.818431 rad and $(x, y, z)$ = (−15 787 554.2, −8 876 334.0, 19 595 952.2) m.

### *G. Step 7 – Geodetic Coordinates of the Sub-Satellite Point*

The native MATLAB function *ecef2lla()* belongs to the Aerospace Toolbox, which is not always installed. An equivalent function with the same name was therefore written (*ecef2lla.m*), based on Bowring's closed-form method [7] on the WGS84 ellipsoid [8] ($a_E$, $b_E = a_E(1-f)$, first and second eccentricities $e_E^2$, $e'^2_E$):

$$p = \sqrt{x^2 + y^2},\ \theta = atan2(z\,a_E, p\,b_E) \quad (16)$$

$$\varphi = atan2(z + e'^2_E b_E \sin^3\theta, p - e_E^2 a_E \cos^3\theta) \quad (17)$$

$$\lambda = atan2(y, x)\,,\; h = \frac{p}{\cos\varphi} - N \tag{18}$$

with $N = a_E/\sqrt{(1 - e_E^2 \sin^2\varphi)}$ the prime-vertical radius of curvature. The sub-satellite point has the same geodetic latitude φ and longitude λ as the satellite and zero altitude. The local function shadows the toolbox one, so the code runs on any MATLAB installation.

### *H. Step 8 – The subsatellite() Function*

Following [1], all the code of Steps 2–7 was removed from *main1.m* and grouped into *subsatellite()*, which calls the previous functions in order (Table III) and returns (λ, φ) in degrees for any *dt*. This makes the propagation of the ground track a single function call per epoch.

TABLE III
CALL CHAIN INSIDE SUBSATELLITE()

| Step | Function | Inputs | Output |
|---|---|---|---|
| 2 | *mean_anomaly* | G, $M_e$, a, $M_0$, dt | M |
| 3 | *eccentric* | e, tol, M | E |
| 4 | *true_anomaly* | e, E | ν |
| 5 | *orbitcoords* | ν, ω, a, e, E | $x_p$, $y_p$ |
| 6 | *satecef* | $\omega_e$, $\Omega_0$, $\dot{\Omega}$, dt, $i_0$, $x_p$, $y_p$ | [x y z] |
| 7 | *ecef2lla* | [x y z] | φ, λ, h |

### *I. Step 9 – Complete Ground Track*

After plotting the sub-satellite point at the target epoch, *main1.m* runs a loop that increases *dt* in steps of $\Delta t = 300$ s (5 min) and plots each new sub-satellite point. The number of steps follows from the repeat condition of the ground track. The orbital period is

$$T = \frac{2\pi}{n} = 43\,077.8\ s \tag{19}$$

i.e. about 11 h 58 min. The ground track closes when the satellite has completed two revolutions, since in that time the Earth has rotated once with respect to the (slowly regressing) orbital plane: $2T = 86\,155.6$ s $\approx 2\pi/(\omega_e - \dot{\Omega}) = 86\,154.6$ s, i.e. about one sidereal day. Hence

$$N_{steps} \geq \frac{2T}{\Delta t} = 287.2 \;\Rightarrow\; N_{steps} = 288 \tag{20}$$

so the loop runs 288 times (24 h); together with the initial point, 289 positions are computed and written to *main1.dat*. Propagating exactly $2T$ returns the satellite to within 0.33 km of the initial sub-satellite point, which confirms that one day of propagation draws the complete ground track.

## IV. RESULTS

### *A. Position of PRN-18 at the Target Epoch*

Table IV summarises the intermediate and final values written to *main1.dat* for the target epoch (angles reduced modulo 2π, as printed by the code). The results were cross verified with an independent re-implementation of the algorithm, with differences below $10^{-6}$ degrees for the 289 epochs.

TABLE IV
RESULTS AT 1 OCT 2026, 10:21:07 UTC

| Quantity | Value |
|---|---|
| *esec* | 382 884.999994 s |
| *dt* | −22 619.000006 s |
| $\Omega_0$ (mod 2π) | 1.168845 rad |
| *M* | 3.919980 rad |
| *E* | 3.915386 rad |
| *ν* | 3.910803 rad |
| $x_p$, $y_p$ | 12 151 505.615 m, 23 756 650.174 m |
| Ω | 2.818431 rad |
| *x* | −15 787 554.194 m |
| *y* | −8 876 334.023 m |
| *z* | 19 595 952.212 m |
| *r* | 26 684 031.1 m |
| λ (long.) | −150.653731° (150.65° W) |
| φ (lat.) | 47.299785° (47.30° N) |
| *h* | 20 317.4 km |

At the target epoch PRN-18 is therefore over the North-East Pacific Ocean, south of the Gulf of Alaska, at 20 317 km above the ellipsoid and moving north-eastwards.

### *B. Ground Track*

Fig. 1 shows the sub-satellite point at the target epoch (labelled PRN-18) and the 288 subsequent positions. From the initial point the satellite reaches its northernmost latitude (55.6° N, over British Columbia) 55 min later, then descends across North America, crosses the equator southbound at 76.9° W over Ecuador (after 3 h 53 min), and reaches 55.6° S over the South Atlantic (6 h 50 min). It crosses the equator northbound at 13.3° E over Gabon (9 h 51 min), reaches 55.6° N over the Urals (12 h 50 min), crosses southbound at 103.1° E over Sumatra (15 h 51 min), reaches 55.6° S south of Tasmania (18 h 50 min) and crosses northbound at 166.7° W over the Central Pacific (21 h 49 min) before closing the track.

The track displays the typical features of a GPS orbit: two lobes per day, ascending (13.3° E, 166.7° W) and descending (76.9° W, 103.1° E) nodes exactly 180° apart, and a maximum latitude equal to the inclination. The geocentric latitude never exceeds $i_0 = 55.57°$, while the plotted geodetic latitude reaches 55.62° because of the ellipsoidal flattening.

### *C. Validation with the FAA NSTB Tool*

The computed position was compared with the real-time WAAS satellite status page of the FAA NSTB [11], captured at the target epoch (Fig. 2). The FAA tool shows PRN-18 over the North-East Pacific, south of the Gulf of Alaska, in good agreement with the computed sub-satellite point (150.65º W, 47.30º N) shown in Fig 1.

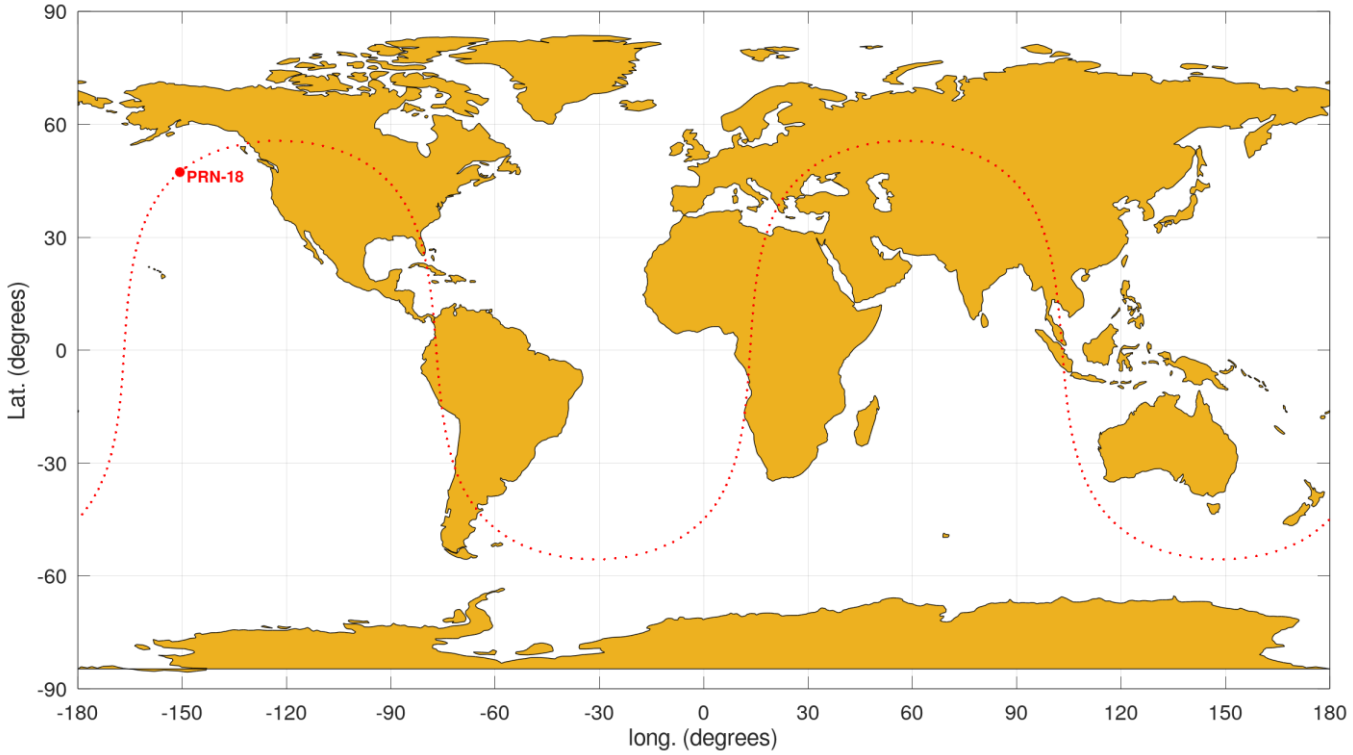



Fig. 1. Ground track of GPS satellite PRN-18 computed with the week-390 almanac (ToA = 405 504 s). The large dot marks the sub-satellite point at the target epoch (1 Oct 2026, 10:21:07 UTC; 150.65° W, 47.30° N); the small dots are the 288 subsequent positions at 5-min intervals (24 h).

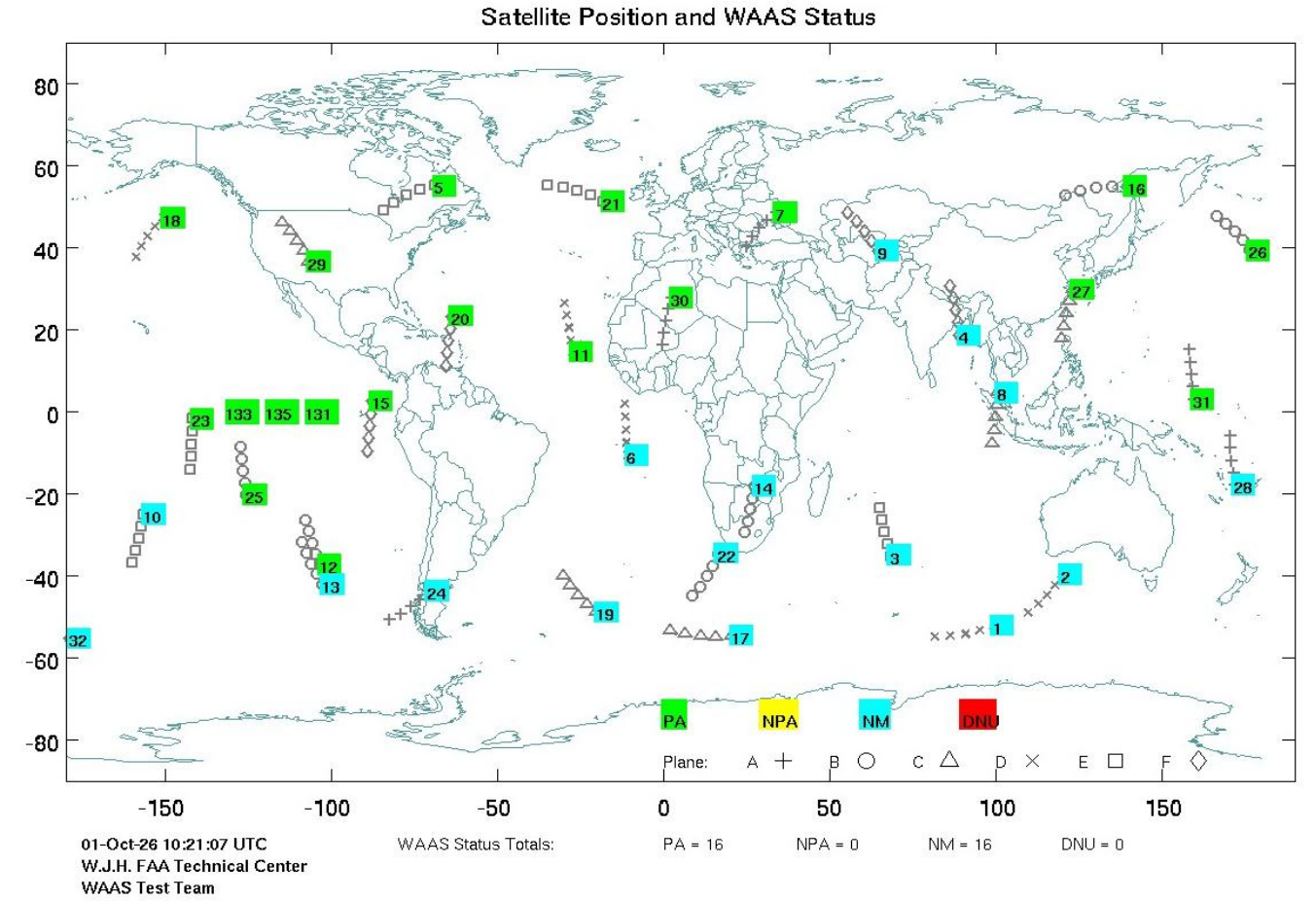



Fig. 2. Screen capture of the FAA NSTB real-time WAAS satellite status tool [11] at the target epoch (1 Oct 2026, 10:21:07 UTC), used as reference to validate the computed position of PRN-18.

### D. Cross-Check with an Earlier Almanac

In a preliminary run the elements of PRN-18 had been taken from an earlier almanac of the same week (ToA = 61 440 s, Sunday 27 September 2026, 17:04 GPS time), which requires propagating the orbit for $dt$ = +321 445 s (3.7 days). This provides an independent cross-check. At the target epoch that almanac places the satellite at 150.83° W, 47.22° N, i.e. 15.7 km from the reference sub-satellite point on the ground and 65.9 km in 3-D (solid line in Fig. 3). The difference stays almost constant over the 24 h.

The origin of this discrepancy is the gravitational parameter. The statement defines $\mu = G\,M_e = 3.985617 \times 10^{14}$ m³/s², whereas the GPS interface specification prescribes $\mu = 3.986005 \times 10^{14}$ m³/s² for the user algorithm [6] (WGS84: $3.986004418 \times 10^{14}$ [8]). The relative difference ($9.7 \times 10^{-5}$) changes the mean motion by $\Delta n = 7.08 \times 10^{-9}$ rad/s, so the along-track error grows as $a\,\Delta n\,|dt|$. That is 4.2 km at the target epoch, up to 11.9 km at the end of the 24-h window (dashed line in Fig. 3) and 60 km for the 3.7-day propagation of the earlier almanac. When both almanacs are propagated with the WGS84 value, they agree within 1.9 km at the target epoch and 3.6 km over the whole day (dotted line), which is the expected consistency between consecutive almanacs.

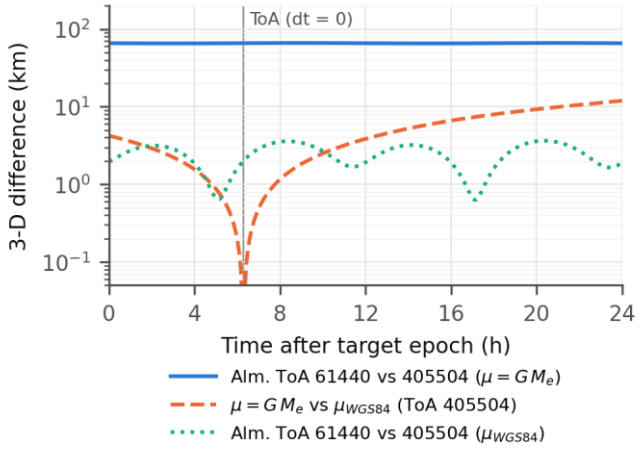



Fig. 3. 3-D position difference over the 24-h window: earlier (ToA 61 440 s) vs. current (ToA 405 504 s) almanac with $\mu = G{\cdot}M_e$ (solid) and with $\mu_{WGS84}$ (dotted), and effect of μ itself on the current almanac (dashed). The vertical line marks the ToA ($dt = 0$).

## V. Discussion

The almanac is a simplified ephemeris: it contains neither the harmonic corrections ($C_{uc}$, $C_{us}$, $C_{rc}$, …) nor the mean-motion correction $\Delta n$ of the broadcast navigation message. Its accuracy is therefore of the order of kilometres, as confirmed by the 2–4 km consistency between consecutive almanacs found in Section IV-D, and it degrades with the time from the ToA. Projected onto the ground, such errors are below about 0.01° in the sub-satellite point, which is negligible for visibility prediction and ground-track plotting but would be unacceptable for positioning.

Two practical consequences arise from the analysis. First, the almanac with the closest ToA should always be used, as *gpstime()* does, to keep $|dt|$ small: with the current almanac $|dt|$ = 6.3 h, against 89 h with the earlier one. Second, the value of μ matters. Replacing $G\,M_e$ by the GPS value would remove a systematic along-track error of about 0.7 km per hour of propagation. This error is invisible at the scale of Fig. 1 but dominates the error budget for propagations of several days.

Time handling is the other critical point. The target epoch must be expressed in UTC (local time in Spain is UTC+2 in summer), and the 18 leap seconds accumulated since 1980 must be added to obtain GPS time. Since the sub-satellite point moves about 2 400–2 800 km along the track every hour, using local summer time instead of UTC (a 2-h error) would displace it by about 4 600 km, so the comparison with the FAA tool would fail.

## VI. Conclusions

A modular MATLAB implementation that computes the sub-satellite point and ground track of any satellite from its Keplerian elements has been developed and applied to GPS PRN-18 with the week-390 YUMA almanac. At 1 October 2026, 10:21:07 UTC, the satellite is located at 150.65° W, 47.30° N, at an altitude of 20 317 km. Newton–Raphson converges in at most three iterations for the near-circular GPS orbit. A loop of 288 steps of 5 min covers two orbital periods (≈ 86 156 s) and draws the complete, closed ground track, which shows the expected two lobes, ±55.6° latitude limits and nodes 180° apart.

The computed position agrees with the FAA NSTB real-time display. The cross-check with an earlier almanac shows that the main source of error is the propagation interval combined with the adopted value of μ. Using the most recent almanac keeps the error at the kilometre level. The same code can be applied to any satellite whose Keplerian elements are known, simply by replacing the almanac data in *main1.m*.

## References


[1] J. Olmos, “Lab. Session 1: Ground track of GPS satellites,” Navegació Aèria, Cartografia i Cosmografia (NACC), EETAC–UPC, Castelldefels, 2025–26, v1.0.

[2] J. Olmos, NACC course slides, EETAC–UPC, Castelldefels, 2025–26.

[3] U.S. Coast Guard Navigation Center, “GPS NANUs, almanacs, & ops advisories.” [Online]. Available: https://www.navcen.uscg.gov/gps-nanus-almanacs-opsadvisories-sof

[4] U.S. Coast Guard Navigation Center, “ICD-GPS-240C: Navstar GPS control segment to user support community interfaces” (YUMA almanac definition). [Online]. Available: https://navcen.uscg.gov/sites/default/files/pdf/gps/ICD_GPS_240C.pdf

[5] CelesTrak, “GPS almanacs (YUMA format).” [Online]. Available: http://celestrak.com/GPS/almanac/Yuma/2026/

[6] Navstar GPS Space Segment/Navigation User Segment Interfaces, IS-GPS-200, GPS Directorate, Los Angeles, CA, USA.

[7] B. R. Bowring, “Transformation from spatial to geographical coordinates,” Survey Review, vol. 23, no. 181, pp. 323–327, 1976.

[8] National Imagery and Mapping Agency, “Department of Defense World Geodetic System 1984: Its definition and relationships with local geodetic systems,” NIMA TR8350.2, 3rd ed., 2000.

[9] J. Sanz Subirana, J. M. Juan Zornoza, and M. Hernández-Pajares, GNSS Data Processing, Vol. I: Fundamentals and Algorithms, ESA TM-23/1. Noordwijk, The Netherlands: ESA Communications, 2013.

[10] P. Misra and P. Enge, Global Positioning System: Signals, Measurements, and Performance, 2nd ed. Lincoln, MA, USA: Ganga-Jamuna Press, 2006.

[11] FAA William J. Hughes Technical Center, “NSTB/WAAS real-time satellite status.” [Online]. Available: https://www.nstb.tc.faa.gov/rt_waassatellitestatus.htm